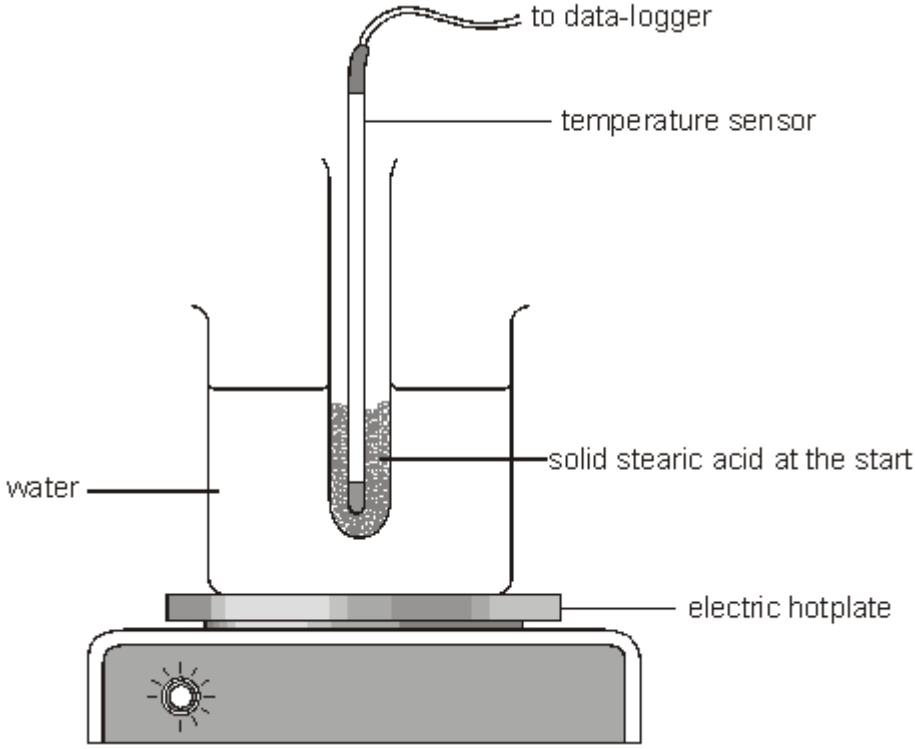
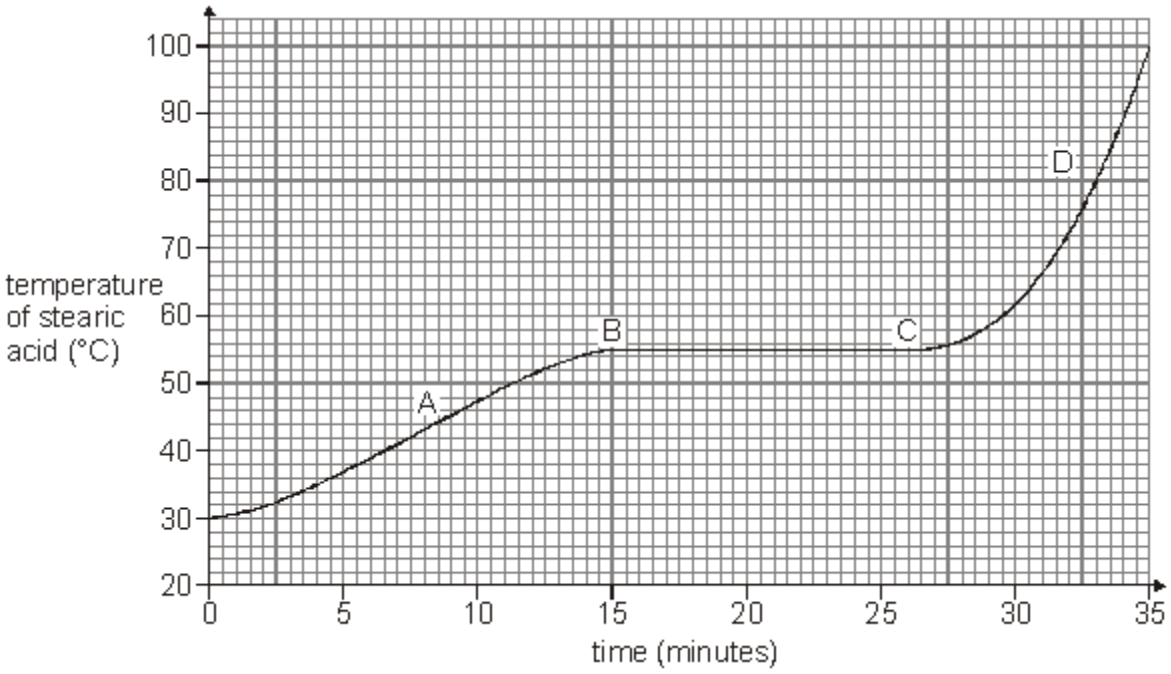
Q1.

Alan put a test-tube containing solid stearic acid into a beaker of cold water.
He heated the water until it boiled.



He used a temperature sensor attached to a data-logger to record the temperature of the stearic acid over a period of 35 minutes.
A graph of the results is shown below.



Stearic acid is a solid at room temperature.

(a) (i) Which **letter** on the graph opposite shows the point at which the stearic acid began to change state?

.....

1 mark

(ii) Use the graph to find the **temperature** at which the stearic acid began to change state.

..... °C

1 mark

(iii) Look at the graph. What was the physical state of the stearic acid:

at point A? .....

at point D? .....

2 marks

(b) The test-tube transfers thermal energy from the water to the stearic acid.

By what method is most of the thermal energy transferred?
Tick the correct box.

conduction	☐	evaporation	☐
convection	☐	radiation	☐

1 mark

(c) Stearic acid boils at 360°C.
The stearic acid could **not** boil in this experiment.
Give the reason for this.

.....

.....

1 mark
maximum 6 marks

Q1.

(a) (i) B

1 (L6)

(ii) 55

accept answers from 54 to 56

1 (L6)

(iii) *point A:* solid

1 (L6)

point D: liquid

1 (L6)

(b) conduction ✓

if more than one box is ticked, award no mark

1 (L5)

(c) any **one** from

- the boiling point of water is less than that of stearic acid
  accept 'the water is not hot enough'
- water could not transfer enough energy
- the maximum temperature of water is 100°C
- it cannot get hot enough
  accept 'stearic acid has a very high boiling point'

1 (L6)

[6]